

Semester project on control and simulation of automation systems

BEng in Robot Systems

4. Semester

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1 Introduction

In life and industry there are a lot of machines that need regulated and precise control, which is why a controller is made to achieve each goal as needed. In robotics controllers can be especially useful when trying to complete difficult tasks with multiple moving parts. Where balance is one of the main hindrances, when trying to design such a system. This project will focus on trying to solve a balance issue of an inverted pendulum.

1.1 Project Goals

With a PLC, motor, conveyor track, inverted pendulum on a cart and a two-degree-of-freedom controller, it is attempted to make the inverted pendulum balance upright using a controller to control the force of the motor. Success is measured by how well the controller balances the pendulum compared to a simulation of the controller, and the best controller is the one whose results are closest to the simulation of said controller.

1.2 Problem Definition

1.3 Constraints & Limitations

The limitations for this project come from the physical setup, and the PLC. The cart is limited to move within two sensors on the conveyor track, and the maximum force for the motor is X. For the PLC it can't run faster than 1000Hz, and sampling via OPCUA is limited to 100Hz.

1.4 System Overview

2 System modeling

2.1 Modelling of the Inverted Pendulum System

2.1.1 System description

The purpose of the modelling process is to obtain a mathematical representation of the inverted pendulum system, which can later be used for simulation and controller design. A mathematical model makes it possible to analyze the dynamic behavior of the system and evaluate different control strategies before implementation on the physical setup.

In this project, two classical modelling approaches were used: the Newton-Euler method and the Euler-Lagrange method. Both methods describe the dynamics of the system, but from different physical perspectives. The Newton-Euler method is based on force and moment balances, while the Euler-Lagrange method is based on the kinetic and potential energy of the system. The resulting nonlinear equations were linearized around the upright equilibrium position to simplify the controller design and implementation in MATLAB/Python.

The physical system consists of a cart with an attached inverted pendulum with an additional weight mounted on top of it. The cart is constrained to move horizontally along a conveyor-driven rail, while the pendulum is free to rotate around its pivot point. The system is actuated by an external force F , which controls the horizontal movement of the cart. The goal of the system is to keep the pendulum balanced in its upright position while controlling the motion of the cart.

The generalized coordinates used to describe the system are:

- x : horizontal cart position
- θ : pendulum angle relative to the vertical axis

The following assumptions were made during the modelling process:

- The pendulum rod is considered rigid
- Friction is assumed to be small and primarily represented by damping coefficients
- The pendulum motion is constrained to a two-dimensional plane
- The system is linearized around the upright equilibrium position

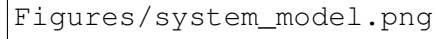
The image is a placeholder for a diagram of an inverted pendulum system. It shows the text 'Figures/system_model.png' centered within a rectangular frame.

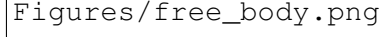
Figure 2.1: Simplified model of the inverted pendulum system showing the generalized coordinates, pendulum angle, and input force.

Table 2.1: System parameters

Parameter	Value
Mass of cart	0.5 [kg]
Mass of rod	0.082 [kg]
Mass of attached weight	0.002 [kg]
Mass of pendulum	0.084 [kg]
Length of rod	0.35 [m]
Length of conveyor belt	1.72 [m]
Radius of belt rollers	0.05 [m]
Gravitational acceleration	9.82 [m/s ²]
Linear damping coefficient	5 [N/(m/s)]
Rotational damping coefficient	0.0012 [Nm/(rad/s)]

2.1.2 Newton-Euler Modelling

The Newton-Euler method is based on applying Newton's second law and rotational moment equations to the system. Free body diagrams were constructed for both the cart and the pendulum to identify the forces and moments acting on the system.



Figures/free_body.png

Figure 2.2: Free body diagram of the inverted pendulum system showing the applied force, gravitational forces, damping effects, and normal force acting on the system.

For the cart, the sum of forces in the horizontal direction was established using Newton's second law:

$$\sum F_x = M\ddot{x} \quad (2.1)$$

The obtained equations contain nonlinear trigonometric terms originating from the pendulum rotation. To simplify the controller design, the equations were linearized around the upright equilibrium position using the small-angle approximations:

$$\sin(\theta) \approx \theta \quad (2.2)$$

$$\cos(\theta) \approx 1 \quad (2.3)$$

Final Linearized Model

$$(I + ml^2)\ddot{\theta} - mgl\theta = ml\ddot{x} \quad (2.4)$$

$$(M + m)\ddot{x} + b\dot{x} - ml\ddot{\theta} = u \quad (2.5)$$

State-Space Representation

$$\dot{x} = Ax + Bu \quad (2.6)$$

$$y = Cx + Du \quad (2.7)$$

2.1.3 Euler-Lagrange Modelling

The Euler-Lagrange method describes the system dynamics using energy relationships instead of direct force balances.

The kinetic energy of the cart is given by:

$$T_{cart} = \frac{1}{2}M\dot{x}^2 \quad (2.8)$$

Using the derived expressions for kinetic and potential energy, the Lagrangian was formulated as:

$$L = T - V \quad (2.9)$$

The equations of motion were then derived using the Euler-Lagrange equation:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = Q_i \quad (2.10)$$

2.1.4 Comparison of the Methods

Both modelling approaches resulted in equivalent dynamic models after linearization, verifying the correctness of the derivations.

The Newton-Euler method provided a more intuitive understanding of the physical forces and moments acting on the system, since it directly uses force and torque balances. However, the method becomes increasingly complex for systems with multiple interconnected bodies.

The Euler-Lagrange method provided a more systematic and compact mathematical formulation by describing the system through energy relationships.

The final linearized model was implemented in MATLAB/Python and used for simulation and controller development. The derived model forms the basis for the PID controllers, pole-placement controller, and the discrete implementation used later in the project.

3 Control theroy

4 PLC implemetation

5 Data Collection

6 Evaluation

7 Discussion

8 Conclusion

References

A Distribution of Tasks

Table A.1: Distribution of Tasks

Task	Noah	Rasmus	Emma	Jannick	Eymundur
Development					
System Modeling					
Control Theroy					
PLC implementation					
Data Collection					
Report					
Abstract					
Introduction					
System Modeling					
Control theroy					
PLC Implementation					
Data Collection					
Results					
Discussion					
Conclusion					

B Code

The code is available in the attached ZIP file as well as on github at:

C Video Showcase

A video of the system running is available at: